

Pattern Arithmetic — Quick Reference, Index and Glossary

Extracted from Pattern_Arithmetic_v15_40 (v15_30 carried the UnCollapse cluster; v15_31 reworded the Mix refusal and UnMix; v15_32 and v15_33 added and tightened composition, structure and contract paragraphs in §1.4; v15_40 writes the bowl reading as $\text{Mix}(B)$, prints the populations behind the reproducibility claims, and prints the enclosure force): section and equation numbers are unchanged. Section numbers (§) and equation numbers (eq.) refer to that document. Every number in Part 9 was recomputed from the paper’s printed inputs (reproduce_v15_40.py).

0 · The soul of the paper, on one page

The soul in eight paragraphs

Thesis (§1). Mathematics need not be performed on numbers alone. Wherever a domain supplies a *unit of truth* — the smallest thing it treats as one — and a small set of rules by which such units combine, obeyed consistently, the unit and its rules form an *arithmetic of that domain*. Counting ($1 + 1 = 2$, “how many?”) is one instance, not the only one: the word “add” names a different rule, and therefore a different kind of unit, for a colour, a note, a word.

The unit (Axiom 1, §2.2). One typed point, not a field: $x = (r, \hat{n}, \chi, \tau)$ with strength $r \in [0, 1]$, direction $\hat{n} \in S^2$ (a *semantic* address, not a place), phase $\chi \in [0, 2\pi)$, sort τ (which buttons may be pressed). The sphere is the stage; the unit is the vector.

The spine (§3.1). A *bowl* of seated units ($\uplus$) is read by a *path* (T) to a *result*; a result may be seated again (*levels are closed*). A unit does not know what it constitutes: **a seated word does not know the poem; the path knows it**. One bowl, several ways to read it.

Six verbs, six questions. $\uplus$ *what is present?* · $\oplus$ *what is combined, nothing discarded?* · $\boxplus$ *what composite results?* · T *what happens, in what order?* · offset *what walk is reused?* · $\bowtie$ *how do two units relate while both still sit?* Three have a counterpart that takes something back ($\setminus$, $\ominus$); Mix’s own, $\boxminus$, is proven impossible.

Six laws worth remembering. $A \uplus A = A$ · $A \oplus A \neq A$ for every A · Mix is lossy and, under the cap, non-associative · $T_a T_b \neq T_b T_a$ yet $T_w^* T_w = \text{id}$ · Assembly never repels ($U_{AB} < 0$) · UnMix does not exist.

What it refuses, on purpose. Mix across differing directions (a claim about $\boxplus$ at one shared direction, not a bar: a two-stage operator is neither adopted nor barred) · UnMix · a geometric reading for any sort with no structure-preserving map (Axiom 0; most physical domains) · a relation across sorts (Collapse holds them side by side and claims none) · treating a costume (storm, forest, table and chair) as a model.

What it keeps honest. Every law is deterministic: the same printed starting units — read at the same time, where any carries an expiry — give the same result. Every choice (the clip, $\alpha = \pi r$, η , δ , the census cut, ϵ_{allow}) is a *named convention*, not a truth of the algebra.

How an atom is counted (§2.10). An atom is N *cores* holding Q informational *vectors*; $D \leq N$ of the cores are distinct. A vector is held by exactly one core, and which core holds which vector (the grouping) is part of the atom. Teaching atom: $N = Q = 1$. Empty atom (black): $N = 1$, $Q = 0$. Beside its content an atom may carry *meta vectors* (time, expiry, place, links, V_{derived}) that describe it and are no part of what it says (§2.11).

1 · The verbs

Verb	Question	Defining rule	Counterpart	Where
Bundle $\uplus$	What is present?	Union on bowls: $B_1 \uplus B_2 = B_1 \cup B_2$. Idempotent, commutative, associative; sort-blind ; discards order and count. $A \uplus A = A$; $A \uplus B = \{A, B\}$.	Remove $\setminus$: $\{A\} \setminus A = \emptyset$ (empty bowl $\neq$ black 0).	§2.5, eq. 5–7

Verb	Question	Defining rule	Counterpart	Where
Collapse $\oplus$	What is combined, with nothing discarded?	Keeps every core of every operand, merges none: $N = \sum N_i$, $Q = \sum Q_i$. Not idempotent: $A \oplus A = A^{(2)}$. Any sorts: each vector carries its own τ_i . Every vector stays in the core that held it; each core keeps its meta vectors.	UnCollapse $\ominus$: takes out one core with every vector it holds, named by an identity key (or by local position when a key matches cores of different content); $N \rightarrow N - 1$ ($N \geq 2$); at $N = 1$ leaves black; Q falls by that core's vector count. Acts along the atom's grain.	§3.3, §3.4
Mix $\boxplus$	What composite state results?	Phasor sum at a shared $\hat{n}$: $z' = \sum r_i e^{i\chi_i}$, $r' = \min(1, z')$, $\chi' = \arg z'$; $\tau' = \tau$; black if $z' = 0$. Over a bowl, one step: $\text{Mix}(B)$, written so it is not read as a fold of $\boxplus$; $z_B = \sum_{x \in B} r_x e^{i\chi_x}$. Over a Collapsed atom the same sum runs over every vector, so repeats add; over a set bowl they count once. Lossy; commutative; binary form non-associative under the cap. Refused across differing $\hat{n}$.	UnMix $\boxminus$: impossible as an inverse of the reading — a Mixed result does not determine its sources. Where the sources were kept, taking one back is Remove or UnCollapse.	§2.6, eq. 8–9; §3.2; §8.2
Transform T	What happens, and in what order?	$T_w(x) = (r_x, \text{Rot}_{\hat{n}_w}(\alpha(r_w)) \hat{n}_x, (\chi_x + \chi_w) \bmod 2\pi, \tau_x)$, $\alpha(r_w) = \pi r_w$. Leaves r_x, τ_x alone. $T_0 = \text{id}$; composition associative; $T_a T_b \neq T_b T_a$.	Inverse actor $w^* = (r_w, -\hat{n}_w, -\chi_w \bmod 2\pi, \tau_w)$, so $T_{w^*} T_w = \text{id}$.	§4.2–4.3, eq. 20, 22
Offset	What walk is reused?	$v = \log_{\hat{n}_m}(\hat{n}_k)$; $\hat{n}_* = \exp_{\hat{n}_w}(\text{transport}_{m \rightarrow w}(v))$; $r_* = r_w$, $\chi_* = (\chi_w + \chi_k - \chi_m) \bmod 2\pi$, $\tau_* = \tau_w$. Copies a walk; does not mint the name. Not T .	—	§5.1, eq. 29–35
Assembly $\bowtie$	How do two units relate while both still sit?	$A \bowtie B = (A, B, \kappa_{AB}, U_{AB})$. Both operands survive unchanged; nothing is fused. $U_{AB} < 0$ always: the bracket $3 + \hat{n}_A \cdot \hat{n}_B$ stays in $[2, 4]$.	—	§2.9, eq. 12–14

Schoolbook + (“how many?”) lives only in **num** mode (decode, add in $\mathbb{Z}$, encode; §6.2). The paper reserves each verb name “the way a keyword is reserved in a programming language”: once given a meaning it is not reused elsewhere for an unrelated or informal sense (Table 1).

2 · The axioms

#	Name	One-line content	Where
0	Admissibility of Sorts	A sort τ may claim a geometric χ - or T -reading only through a map $\varphi_\tau : D_\tau \rightarrow \mathcal{S}$ with $\varphi_\tau(a \circ_\tau b) \approx \boxplus(\varphi_\tau(a), \varphi_\tau(b))$ or $T_{\varphi_\tau(a)}(\varphi_\tau(b))$ at a stated fidelity (Tier 1). Tier 0 (word , num) is inert. An inadmissible sort (e.g. snowflake, D_{6h} growth) may still be seated, at Tier 0 only. A sort absent from the table is undecided, not admitted.	§2.1, eq. 1
1	Normalized Unit Truth State	A pattern is one typed point, $x = (r, \hat{n}, \chi, \tau)$, $r \leq 1$ a normalized activation (not a probability). A bowl is a separate finite set of such atoms. The bound is imposed by clipping — a choice, with its trade stated (not injective; $r = 1$ is a value, not a limit).	§2.2, eq. 2
2	Absolute Null State (Black)	$r = 0 \Rightarrow x = \mathbf{0}$: direction and phase unused — an <i>empty seat</i> . As an atom: $N = 1$, $Q = 0$. A vector of strength zero is not an arrow.	§2.3, eq. 3
3	Saturation (White)	The upper pole of the same bounded stage: one full-spectrum name, or a saturated bowl. No unit is $r = \infty$. “Density” = many identities share the stage without overwrite.	§2.4, eq. 4
4	Bundle and Remove	$\uplus$ collects (union, idempotent); $\setminus$ removes one member. Sort-blind: sort discipline governs what may later be <i>done</i> with a bowl, not who may sit in it.	§2.5, eq. 5–7

#	Name	One-line content	Where
5	Mix as a reading	A phasor operator: combines χ among states that already share $\hat{n}$; never moves an arrow. A reading returns a state and leaves the bowl intact.	§2.6, eq. 8–9
—	Mix across differing directions: refused; chladni inclusion	Cross-mode coexistence is $\uplus$, not $\boxplus$: mode space is not closed under addition. Refusal is a claim about $\boxplus$ at one shared direction, not a permanent bar: rotating onto a named axis is T then Mix, and a two-stage operator (lift to a higher space, then read) is neither adopted nor barred. chladni is Tier 1: exact within one mode, undefined across modes.	§2.7, eq. 10
—	Rotation keeps $\hat{n}$ on the sphere	$\mathbf{R} \in \text{SO}(3)$ turns $\hat{n}$ and leaves r, χ, τ : the geometric engine of T , not a second blend.	§2.8
6	Assembly	Relates two intact units by a contact point and an interaction energy; never repels; completed by the enclosure’s seats $\vec{P}$.	§2.9, eq. 12–14
7	Informational vectors and derived per-atom metrics	An atom may hold N cores of Q informational vectors under one closure, each an instance of Axiom 1 with its own sort τ_i ; a vector is held by exactly one core, recorded by a containment vector. Defines V_{derived} , seven scalar metrics, Q_{eff} , the empty atom.	§2.10
—	Meta vectors (not an axiom)	What an atom carries <i>beside</i> its content: uncapped, not informational, in neither N nor Q ; rides with its core.	§2.11

3 · The unit, its coordinates, and the counts

Symbol	Domain	What it answers
r	$[0, 1]$	How strong / how confidently held is this unit?
$\hat{n}$ (or (θ, ϕ))	S^2 (higher: $S^{N_{\text{dim}}-1}$)	What direction — the semantic address?
χ	$[0, 2\pi)$	What internal phase / mood / hue?
τ	$\{\text{word, num, color, note-pitch, chladni, ...}\}$	What sort — which operators may legally act on it? (<i>au</i> does not move the arrow.)

Term	Meaning
teaching atom	The 4-tuple of Axiom 1: one arrow, $N = Q = 1$. One state among several an atom can be in.
atom (Collapsed)	N cores holding Q informational vectors $v_i = (r_i, \hat{n}_i, \chi_i)$, $0 < r_i \leq 1$, under one closure; each vector carries its own sort τ_i . Arises by Collapse or is seated directly by whatever supplies the seat.
core	A group of vectors held together in an atom. Holds any number of vectors: one, many, or none. A vector is held by exactly one core, as a nucleus holds its electrons; no verb moves a vector to another core. “Core” is reserved for this sense.
containment vector	For each vector, the position of its core in the atom’s list of cores: the grouping itself. Part of the atom’s value, so equality compares it; not a meta vector.
grain	The partition of an atom’s vectors into cores: fine when each core holds one reading, coarse when a core holds a window. UnCollapse acts along it. Fixed at seating, coarsened by seating again (many-to-one), refined by no verb.
identity key	A core’s name: the tuple of meta vectors it already carries (sort, source link, timestamp); no new kind; never enters equality.
local position	A core’s place in its atom’s list of cores, from 0; unique inside the atom, meaningless outside. Collapse shifts the second atom’s positions by the first’s N ; UnCollapse lowers positions above the one removed. The fallback when a key matches cores of different content.
N	Number of cores . A count of the stored atom: does not change with the reading time t .
Q	Number of informational vectors — “the atom’s info number”. $Q \geq N$ is usual, not a rule. Read over the cores valid at t .
D	Number of distinct cores among the N : two cores coincide exactly when $\uplus$ would seat them together (content only; empty cores coincide; meta vectors never enter). $D \leq N$, equality iff every core is distinct. Stored count.
$N - D$	How many times some core repeated.
$A^{(k)}$	k identical copies of A ; has kN_A cores. $A \oplus A = A^{(2)}$.

Term	Meaning
empty atom	Black: $N = 1, Q = 0$. $\mathcal{I}_{\text{str}}, \mathcal{I}_{\text{vol}}, \mathcal{I}_{\text{en}}, V_{\text{derived}} = 0$; $\mathcal{I}_{\text{mean}} = 0, \mathcal{I}_{\text{vcon}} = 0$ by convention; $\mathcal{I}_{\text{agr}}, \mathcal{I}_{\text{scon}}, Q_{\text{eff}}$ undefined. Collapse keeps a silent contribution: $A \oplus 0$ has one more core, same vectors; $0 \oplus 0 \neq 0$.
mass number / atomic number	N reads like a nucleus's mass number (every core counted), D like its atomic number (distinct cores). An analogy of shape only.

4 · Metrics (Axiom 7, §2.10)

All seven metrics are written $\mathcal{I}$ (script I, the information atom) with a subscript: str, vol, mean, en, agr, vcon, scon. Named in the paper: *atom strength*, *atom volume*, *mean vector volume*, *atom coherence*, *effective vector count*; $\mathcal{I}_{\text{en}}$ is the atom's *information energy*; $\mathcal{I}_{\text{agr}}, \mathcal{I}_{\text{vcon}}, \mathcal{I}_{\text{scon}}$ are its fifth, sixth and seventh metrics: *atom agreement*, *volume concentration* and *shell concentration*. All presuppose an atom of Q informational vectors $v_i = r_i \hat{n}_i$. None is used by any verb; each is read over the cores valid at the reading time. Identity: $|V_{\text{derived}}|^2 = \mathcal{I}_{\text{en}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)$, the structure of a variance of a sum.

Symbol	Formula	What it measures	Requires
V_{derived}	$\sum_{i=1}^Q r_i \hat{n}_i$ (χ excluded)	The raw, uncapped vector sum; per core and per atom (the atom's is the sum of its cores'). Reads as an effective pair $r = V $, $\hat{n} = V/r$ for a weight or a direction; not itself a unit; may act as T 's actor as it is (uncapped, angle $\pi V \bmod 360^\circ$; it follows size, not content).	any Q
$\mathcal{I}_{\text{str}}$	$\sum_i r_i$	Atom strength . Denominator of atom coherence.	any Q
$\mathcal{I}_{\text{vol}}$	$\sum_i r_i^3$	Atom volume . Strictly $< \mathcal{I}_{\text{str}}$ whenever any $r_i < 1$ — forced by the cap.	any Q
$\mathcal{I}_{\text{mean}}$	$\mathcal{I}_{\text{vol}}/Q$	Mean vector volume .	$Q \geq 1$ ($:= 0$ at $Q = 0$)
$\mathcal{I}_{\text{en}}$	$\sum_i r_i^2$ (eq. 15)	Information energy . Stands on its own; the energy the vectors would carry if every pair were orthogonal.	any Q
$\mathcal{I}_{\text{agr}}$	$\frac{\sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)}{\sum_{i < j} r_i r_j}$ (eq. 16)	<i>Atom agreement</i> (fifth metric). Strength-weighted mean cosine similarity across all internal pairs; separates active cancellation (< 0) from mere non-reinforcement, which atom coherence cannot.	$Q \geq 2$
$\mathcal{I}_{\text{vcon}}$	$\frac{Q^2}{Q-1} \text{Var}(s_1, \dots, s_Q)$, $s_i = r_i^3 / \mathcal{I}_{\text{vol}}$ (eq. 17)	<i>Volume concentration</i> (sixth metric). 0 when even, 1 when one vector holds all. Equals the normalized Herfindahl index of the cubed shares.	$Q \geq 2$ ($:= 0$ at $Q \leq 1$)
$\mathcal{I}_{\text{scon}}(w)$	1 if $K = 1$; else $\frac{K^2}{K-1} \text{Var}(p_1, \dots, p_K)$, $p_k = n_k/Q$, bins of width w (default 0.1) (eq. 18)	<i>Shell concentration</i> (seventh metric; across shells). Catches what $\mathcal{I}_{\text{vcon}}$ cannot: many vectors independently landing at the same magnitude.	$Q \geq 1$
Q_{eff}	$\mathcal{I}_{\text{str}}^2 / \mathcal{I}_{\text{en}}$	Effective vector count — not an eighth metric. $1 \leq Q_{\text{eff}} \leq Q$; scale-free; the largest vector holds at least $1/Q_{\text{eff}}$ of the strength; a <i>count</i> , with Q_{eff}/Q the size-free form.	$\mathcal{I}_{\text{str}} > 0$
atom coherence	$ V_{\text{derived}} / \mathcal{I}_{\text{str}}$	How much strength survives into one resultant; 1 for aligned vectors, falling toward 0. Also a normalization of V_{derived} into $[0, 1]$.	$\mathcal{I}_{\text{str}} > 0$

5 · Meta vectors (§2.11)

What they are. Values that describe an atom and are no part of what it says. Not capped; not informational; counted in neither N nor Q ; outside the seven metrics and D ; no admissibility of their own; **not a coordinate** (the four coordinates and the sort tag are unchanged). They ride with the cores they describe. Anything beyond that — what a meta vector *means* — is extracted by context or by the path that reads it.

Kind	Held	What it is
V_{derived}	per core and per atom	Sum of the informational vectors, recomputed whenever the content is. Already in the paper (Axiom 7).
seat $\vec{P}$	per unit or bowl	External position in unbounded Euclidean space; enters only $d_{AB} = \vec{P}_B - \vec{P}_A $ (§10.1). Already in the paper.
$V_{\text{timestamp}}$	per core	When the core was born, an absolute (non-cyclic) time, so age = $t - V_{\text{timestamp}}$ is derived, never stored. An optional clock-face vector (minute on ϕ , hour on χ , θ at the equator) groups by time of day but cannot replace it. A Collapse result also gets its own new timestamp.
V_{expiry}	per core	When the core stops being valid. Absence means no expiry. Valid at t iff t is earlier than every expiry it carries.
V_{location}	per core, and per atom where a use supplies one	A point on a sphere (latitude as polar angle, longitude as azimuth). Not a sort: a place is a coordinate, not a claim with a strength.
V_{link}	per core	Names another atom by one of that atom’s meta vectors, never by content (a word is never linked to a poem). Only backward links are stored, at birth; forward and bidirectional links are views (a search for atoms whose backward link names a given one). A Collapse result gets a backward link naming its sources.
V_{identity}	per core	The identity key: the tuple (sort, source link, timestamp) of meta vectors the core already carries; no axis of its own; compared by equality; never enters the equality of atoms.
$V_{\text{container}}$	per vector	The containment vector: the position of the vector’s core in the atom’s list of cores; $\mathbb{R}^1$, an index. Listed beside the meta vectors but enters the equality of atoms, being the grouping itself.

Rules that govern all of them

Rule	Statement
Form	Always vectors, with no constraint on form. A scalar meta vector takes a designated axis of its core and carries its value as the coordinate there (uncapped), so vector distance is the difference of values; more coordinates (a plane or more) where a kind needs them. Which axis suits which kind is a recommendation (a catalogue, §6.7), not an axiom.
Turning	Turn an atom as a whole (same rotation and phase shift on each vector): V_{derived} turns with it; scalar meta vectors are carried unchanged; angles, magnitudes and distances among the information vectors and V_{derived} are preserved.
Atoms are not edited	An atom is a value. Change is Collapse or UnCollapse, each returning a new atom; recorded meta vectors never change after birth. There is no “last modified”.
Equality	Two tests. <i>Equal</i> : the same vectors in the same grouping; meta vectors never enter (what $\uplus$, D and UnCollapse use). <i>The same individual</i> : equal, and the meta vectors agree too ($=$ against “is”). Equality is exact; measured readings are made comparable when they are seated. Were the timestamp part of the value, $N - D$ would always be 0 and $A \uplus A = A$ would never apply.
Identity and containment	Identity is read from meta vectors (the identity key) and never enters equality; it can be dropped and UnCollapse still works by position. Containment is <i>not</i> a meta vector: it is the grouping, part of the atom’s value.
Results	Every result is a new atom with its own new timestamp and a backward link naming its operands; the cores that survive keep theirs. A result is also one new core on top of those: it carries its own V_{derived} and may carry its own V_{location} and V_{expiry} , supplied by a use; no verb computes them from the operands’.
Expiry is a reading	Everything is read at a time t (default now). An expired core contributes no vectors and reads as a black core; N , D are stored counts and do not change with t ; Q , V_{derived} , the metrics and every process read only valid cores. An atom’s expiry applies to all its cores; a contribution’s expiry is carried onto each of its cores at Collapse. An expired core is noise: it and its vectors may be removed, by UnCollapse selected by expiry (a new atom; N and D then fall). A backward link is lineage, not participation, and stays readable.
Nothing loses identity	Under Collapse, Bundle and Mix the individual persists as meta vectors; this is persistence, not recovery. Mix leaves its sources unconsumed and names them by backward links; UnMix stays impossible.
Absence is not zero	For any meta vector: an absent location is not $(0, 0)$; “expiry 0 means none” would read as already expired.

Rule	Statement
No verb changes one	No verb changes a meta vector or takes one as an operand, except that T 's actor may be V_{derived} , read as it is; a verb may read one (Assembly reads V_{derived} and the seat).
Uses	Group atoms for a task (collect, order, assemble). Grouping needs a comparison stated with each kind (great-circle for a location, cyclic for a clock face, equality for an identifier); none is common to all. Direction is the cleaner assembly rule; magnitude enters as a weight. Links declare structure and Assembly's energy produces it from geometry and a seat; the two are separate. Links can carry an order past $\uplus$, which does not read them, so order still comes from outside and the path is where it is walked. At contact the best partner of an atom of strength 0.60 has strength 0.30; at separation 3 it is the strongest available.

6 · Other named quantities, by operator

Symbol	Formula	Lives in	What it is
U_{AB}	$-\frac{r_A r_B}{2} \cdot \frac{3 + \hat{n}_A \cdot \hat{n}_B}{(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}}$ (eq. 13)	§2.9	Assembly's interaction energy. Product $r_A r_B$: how strong; bracket in $[2, 4]$: how aligned (at most a factor of two); denominator: how far. Always < 0 for nonzero strengths; exactly 0 for a black unit.
ε_{AB}	$r_A + r_B$	§2.9	Softening length: a convention that reads r as a radius.
κ_{AB}	$\hat{n}_A + \hat{n}_B$ (unnormalized; eq. 14)	§2.9	Contact point; defined at every angle, = $\mathbf{0}$ at antipodal. Its magnitude falls from 2 to 0.
polarity	$ r_A - r_B $	§2.9	Labels a bond as balanced or lopsided; does not change what the relation does.
$d_{AB}, \hat{r}_{AB}$	$ \vec{P}_B - \vec{P}_A , (\vec{P}_B - \vec{P}_A)/d_{AB}$ (eq. 46)	§10.1	Separation and separation axis; no axis derived from orientation is used.
$\alpha(r_w)$	πr_w	§4.2	Strength to rotation angle; $r_w = 1$ gives 180° .
z_B	$\sum_{x \in B} r_x e^{i\chi_x}$; Mix(B) = ($\min(1, z_B)$), $\hat{n}$, $\arg z_B$, τ) (eq. 9)	§2.6	Mix's one-step bowl reading; the reason binary Mix is non-associative under the cap.
$\log_{\hat{p}}, \exp_{\hat{p}}$	$\frac{\theta}{\sin \theta}(\hat{q} - (\hat{p} \cdot \hat{q})\hat{p})$; $\cos\ v\ \hat{p} + \sin\ v\ v/\ v\ $ (eq. 29–30)	§5.1	Walk from p to q as a tangent vector, and back.
transport	$\text{Rot}_{\hat{u}}(\theta_{mw}) v$, $\hat{u} = \hat{n}_m \times \hat{n}_w / \ \cdot\ $ (eq. 32–33); general N_{dim} : a rotation within the plane the two seats span (eq. 35)	§5.1	Carries the walk from seat m to seat w . The two seats name the turning plane, so no Collapse is needed. Exactly the cross-product form at $N_{\text{dim}} = 3$.
F_i	$-\nabla_{\vec{P}_i} \sum_{j \neq i} U_{ij}$; step $\vec{P}_i \leftarrow \vec{P}_i + \delta F_i$ (eq. 47)	§10.2	Force on a seated unit, descended with no mass and no momentum.
freeze at resolution η	$\max_i F_i < \eta$ (eq. 48)	§10.2	A stated checkpoint, not exact rest: the pull is positive at every finite separation, so left longer every configuration drifts toward one pile (proved for the continuous flow with a finite population of fixed strengths; observed for the Euler step).
camp, body $\vec{P}_{c,k}$	$\frac{\sum_{i \in G_k} r_i \vec{P}_i}{\sum_{i \in G_k} r_i}$ (eq. 49)	§10.3	A frozen group and its strength-weighted centre, read <i>after</i> motion. A system of camps is the <i>set</i> of bodies, never one grand average.
ϵ_{allow}	named, not derived	§10.3	Checkpoints whose bodies differ by less are read as the same freeze.

7 · Sorts, tiers, and who may do what

Sort	Native law $\circ_\tau$	Mix/ T map?	Tier
num	schoolbook +	no (chart only)	0

Sort	Native law $\circ_\tau$	Mix/ T map?	Tier
word	none claimed	n/a	0
note-pitch	pitch-class sum	near-exact	1
color	additive light	phasor Mix (here; a toy $0^\circ/120^\circ/60^\circ$ chart — not CIE, not linear RGB)	1
chladni	modal superposition	exact within a mode; undefined across	1 (restricted)
snowflake	D_{6h} growth	no	Tier 1 no; Tier 0 yes

Pair	+	$\uplus$	$\oplus$	$\boxplus$	$T_w(x)$	$x - a + b$
(word,word)	no	yes	yes	yes [†]	yes	yes
(num,num)	yes	yes	yes	no (clock trap)	no	no
(word,num), (num,word)	no	yes	yes	no	typed	no
(color,color)	no	yes	yes	yes	yes	open
(note-pitch,note-pitch)	no	yes	yes	yes	yes	open
(chladni,chladni)	no	yes	yes	yes (within a mode)	open	open
any other pair	no	yes	yes	open	open	open

§6.4 (Table 3). Every Mix entry is also subject to the place restriction (shared $\hat{n}$). $\uplus$ is *yes* for every pair: a bowl records presence and makes no domain claim. $\oplus$: *yes* for every pair, like $\uplus$, and it keeps the count. [†]Mix on two words is geometry, not a χ -reading of language. **A document is a sequence, not a blend** (§6.3): words and numbers meet only by *juxtaposition*, *typed action* (T across sorts needs an explicit output sort) or *cast*; dawn $\uplus$ 37.4 has no meaning until one side is cast.

8 · Dimensions: how an actor grows beyond three

Space	Sphere	Fixed part (the “axis”)	What turns
$\mathbb{R}^2$	S^1	a point, dim 0	one plane
$\mathbb{R}^3$	S^2	a line, dim 1	one plane
$\mathbb{R}^4$	S^3	a plane, dim 2	one plane
$\mathbb{R}^5$	S^4	a 3-D space, dim 3	one plane
$\mathbb{R}^{N_{\text{dim}}}$	$S^{N_{\text{dim}}-1}$	dim $N_{\text{dim}} - 2$	one plane

A teaching atom ($Q = 1$) is a legal actor ****only at $N_{\text{dim}} = 3$ ****; beyond that the actor’s atom must supply $N_{\text{dim}} - 2$ mutually orthogonal arrows ($Q \geq N_{\text{dim}} - 2$ is necessary, not sufficient): $T_w(x) = (r_x, \hat{n}_x^F + \text{Rot}_{\alpha(r_w)} \hat{n}_x^P, \chi_x + \chi_w, \tau_x)$, eq. 23. Orientation of $F^\perp$ is a convention, stated once. Checked invertible at every angle tested and non-commutative at $N_{\text{dim}} = 4, 5, 8, 12, 20, 50$; exact at $N_{\text{dim}} = 3$. Offset’s transport generalizes without any Collapse. Mix, Bundle and $\mathcal{I}_{\text{agr}}$ are dimension-blind. The paper keeps $\mathbb{R}^3$ as its standing assumption (§4.4).

9 · Numbers worth remembering (all recomputed from the printed inputs)

Claim	Printed value	Where
$(1\angle 0^\circ) \boxplus (1\angle 120^\circ)$ (red, green)	$1\angle 60^\circ$, yellow	§2.6
$0.6\angle 0^\circ \boxplus 0.8\angle 120^\circ$	$r' \approx 0.721, \chi' \approx 73.9^\circ$	§2.6, §8.1
$A \boxplus A$	$r' = \min(1, 2r)$ — not idempotent	§2.6
$A = B = 1\angle 0^\circ, C = 1\angle 180^\circ$ (by hand)	$(A \boxplus B) \boxplus C = \text{black}, A \boxplus (B \boxplus C) = 1\angle 0^\circ$; one-step $z_B = 1$	§2.6
Clock, $M = 24$: 7 + 5 and Mix of the two marks	12 (decode, add, encode); Mix gives $\chi' = 90^\circ$, hour 6	§6.2
Sun, Wind, Sky ($r_{\text{Sun}} = 1, r_{\text{Wind}} = 0.7$)	Sun then Wind $(-0.866, -0.500, 0.011)$; Wind then Sun $(0.278, -0.500, 0.820)$; $\alpha_{\text{Wind}} = 126^\circ$	§4.6

Claim	Printed value	Where
Four-dimensional actor, $r_w = 0.5$	$T_w(\hat{n}_x) = (\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{\sqrt{2}})$, recovered by w^*	§4.7
Order gap, $N_{\text{dim}} = 4, 5, 8, 12, 20, 50$	0.683, 0.688, 0.649, 0.584, 0.489, 0.331	§4.4
Storm: Deepen / Clear endpoints	$(0.7071, -0.6124, -0.3536)$, $(110.7^\circ, 319.1^\circ)$ / $(0.3536, -0.6124, -0.7071)$, $(135^\circ, 300^\circ)$; both $\chi = 135^\circ$	§9.1
Offset on an equator $(0^\circ, 30^\circ, 90^\circ)$	lands at $(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$, 120° — a protractor check, not the operator	§5.2
Unmixing: $0^\circ/120^\circ$ against $10^\circ/100^\circ$	same phasor sum (weights 0.643, 0.766); residual at rounding level	§8.2
Alignment, $r = 0.85$, pairs 5.7° apart	$Q = 2$: raw 0.72, $\alpha = 0.995$; $Q = 5$ on a cone of half-angle 5.7° : 7.14, 0.988	§2.10
Three aligned at $r = 0.5$	$ V = 1.5$, a turn of 270° as T 's actor, read as it is; $51.877 \rightarrow 337.9^\circ$, $5187.748 \rightarrow 314.6^\circ$	§2.10
Dominant vector 0.95 against three at 0.2	volume share 0.973	§2.10
Best partner for an atom of $r = 0.60$ (Assembly energy)	at contact $r_B = 0.30$; at separation 3 the strongest available (1.0)	§2.11
$\mathcal{I}_{\text{scn}}$: 10@0.98, 3@0.3, 400@0.65	0.908 at $w = 0.1$ and $w = 0.05$	§2.10
Enclosure, six printed units, $\eta = 10^{-5}$	628, 1261, 2526, 6320 steps at $\delta = 0.02, 0.01, 0.005, 0.002$; two camps $\{1, 2, 3\}$, $\{4, 5, 6\}$; bodies $(0.33336, 0.33333, 0)$, $(40.33330, 0.33333, 0)$; runs agree to 2×10^{-7} ; r -weighted and plain bodies differ by 2.4×10^{-8}	§10.4
Enclosure, pair $r = 0.9, r = 0.1$, $\delta = 0.02$	separation never increases, $3 \rightarrow 1.80$; dividing each step by r^3 (a 729-fold weighting) makes it oscillate	§10.4

What reproduces, and what is sensitive: every printed number is recomputed by `reproduce.v15.40.py` from its printed inputs. One count, the number of steps on which the weighted pair's separation increases, depends on rounding and is printed as a range (830 to 855 of 3000). Shell bins are defined in exact arithmetic, because $\lfloor 0.3/0.1 \rfloor$ is 2 in floating point.

10 · Named conventions — choices, not truths

Convention	What it fixes	Where
the clip $r' = \min(1, z)$	How the cap is imposed. An order-preserving bijection $ z \mapsto z /(1 + z)$ would bound without discarding but makes $r = 1$ a limit. The clip is kept; the trade is stated.	§2.2
$\alpha(r) = \pi r$	The simplest monotone map from strength to rotation angle.	§4.2
orientation of $F^\perp$	Which direction counts as $+\alpha$ beyond three dimensions; any consistent choice, stated once.	§4.4
toy colour chart	Phasor Mix on $0^\circ/120^\circ/60^\circ$; neither CIE nor linear RGB.	§2.1, §2.6
numeric chart	$\chi = 2\pi k/M$, $M = 24$, at a fixed direction with $r = 1$.	§6.1–6.2
shell width w	Default 0.1 (ten shells on $[0, 1]$); reported with $\mathcal{I}_{\text{scn}}$.	§2.10
η , δ , census cut, ϵ_{allow}	Freeze threshold; step size; how camps are counted; when two checkpoints are the same freeze. The enclosure's four measurement conventions.	§10.2–10.3
reading time t	Default now; fixing it keeps every law deterministic. “Now” is external context, as the lexicon is.	§1.4, §2.11
standard dictionaries and the meta-vector catalogue	Recommendations, recorded and left open; not axioms.	§6.7

11 · Stated open, or not claimed

Item	Status in the paper	Where
Collapsed atom as the actor of T	V_{derived} acts as it is (angle mod 360°); a worked example is not claimed here. A bowl is not an actor: T takes one unit.	§2.10

Item	Status in the paper	Where
Reducing a path by merging close actors	Sequential actors already compose into one rotation for free (closure). Replacing merely <i>close</i> actors is a different operation, coarse-graining, whose shape only is stated: it yields an additional derived actor <i>beside</i> its sources and removes none. Which merge rule, and what drift bound, belongs to Inference.	§12.1
Dimensionality	The sphere S^2 is where the idea is taught; embedding-scale dimensions are a later choice.	§6.8, §12.1
Offset on Tier-1 sorts; antipodal offset	Untested here; antipodal seats have no unique short geodesic and that convention is not used.	§5.1, §6.4
A relation across sorts	No rule stated at all (a colour with a charge; a chord with a valence bond); Collapse only holds them side by side and claims none.	§1, §2.1, §3.3
Verdicts	Saying what a resting state <i>means</i> is a later reading, not a verb here.	§1

12 · Document map

§	Title	What you will find
1	Introduction	Thesis; examples of other arithmetics; Pāṇini; scope of the claim (1.4), with determinism, composition, what kind of structure this is, and what the algebra promises and a use supplies.
2	Axiomatic Foundations	Axioms 0–7, Mix refusal and chladni (2.7), rotation (2.8), Assembly (2.9), metrics (2.10), meta vectors (2.11).
3	Process Principles	Bundle–path–result and levels (3.1); UnMix (3.2); Collapse (3.3); UnCollapse (3.4).
4	Transformation Algebra	T : definition, properties, beyond three dimensions, poems as trajectories, worked examples.
5	Offset	Log, carry, exponential; the equator check; offset is not T .
6	Encodings, Sorts, Mixed Documents	num and word ; the clock; a document is a sequence; operator legality; lexicons; standard dictionaries.
7	A Sketch of Conventions	Case and language do not move the address; a bag is not a sentence; a numeral is not a word until cast.
8	Operational Algebra	Mix of two shells; why unmixing is ill-posed; spectral storage (later).
9	Higher-order systems	Costumes: storm, table and chair, forest, pathogen.
10	The Enclosure	Seats, rest and freeze, camps and bodies, a replayable configuration, what it does not supply.
11–12	Discussion; Conclusion	Related work (Bloch sphere, embeddings, transformers, RotatE, von Mises–Fisher, Riemann sphere, $SO(3)$, spherical harmonics); future work.

13 · A–Z index of terms with special meanings

Term	Meaning	Where
Actor w	The unit that acts in $T_w(x)$: supplies axis, strength ($\rightarrow$ angle) and phase. Beyond three dimensions it must supply $N_{\text{dim}} - 2$ orthonormal arrows.	§4.2, §4.4
Admissible (geometrically)	A sort with a stated map φ_τ at a stated fidelity (Tier 1). Absent from the table = undecided, not admitted.	§2.1, eq. 1
Age	$t - V_{\text{timestamp}}$; derived, never stored (a stored age is wrong once time passes).	§2.11
Allowance ϵ_{allow}	Checkpoints whose bodies differ by less are read as one freeze; a named convention.	§10.3
Arithmetic (of a domain)	A unit of truth plus a small set of rules for combining such units, obeyed consistently.	§1
Arithmetic (this paper’s sense)	A unit, the operations on it, and the properties each operation is stated and checked to have.	§1.4
Assembly $\bowtie$	Relates two intact units by (κ_{AB}, U_{AB}) without fusing them.	§2.9

Term	Meaning	Where
Atom	The typed unit. <i>Teaching atom</i> : one arrow. <i>Collapsed atom</i> : N cores, Q vectors, one closure.	§2.2, §2.10
Atom coherence	$ V_{\text{derived}} /\mathcal{I}_{\text{str}}$. With the direction of V_{derived} it is a colour's locator: colour lives in $\hat{n}$, phase is kept for mixing, and $ V_{\text{derived}} $ measures size, not colour.	§2.10
Axis ($\hat{n}_w$, fixed part F)	The line (3-D) or subspace of dimension $N_{\text{dim}} - 2$ that T leaves fixed.	§4.4
Black 0	$r = 0$: an empty seat; $N = 1$, $Q = 0$; $T_0 = \text{id}$; the identity of nothing else. Distinct from the empty bowl $\emptyset$.	§2.3, §2.5
Body $\vec{P}_{c,k}$	A frozen camp's strength-weighted centre, measured afterwards by an observer; a description, not a destination.	§10.3, eq. 49
Bowl	A finite set of seated units: who is present. No order, no reading, no representation of the reading to come. May carry meta vectors; $\uplus$ does not read them.	§2.2, §2.5, §3.1
Bundle $\uplus$	Union on bowls; idempotent; sort-blind.	§2.5
Camp	A separated group frozen at resolution η ; their number is whatever the law, the seats and η produce.	§10.3
Cast (two uses)	(a) lexicon cast E_L : token to state (eq. 42); (b) sort cast, e.g. $E_{\text{num} \rightarrow \text{word}}$ (§6.3). Normalizing V_{derived} into $[0, 1]$, to seat it as a unit or read atom coherence, is a normalization, not a cast (§2.10).	§6.6, §6.3, §2.10
Census cut	The stated separation (or chain of short links) used to count camps; a convention.	§10.3
Chart	An exact, invertible encoding (e.g. num); not a Mix or T homomorphism.	§2.1, §6.2
Checkpoint	One freeze read at some η after some time; not a destination.	§10.3
Clip	$r' = \min(1, z)$; the chosen way to bound strength; not injective.	§2.2, §2.6
Clock trap	Mix on (num , num): geometrically defined, type-refused. 7 and 5 Mix to hour 6.	§6.2
Collapse $\oplus$	Count-preserving combination of atoms of any sorts; keeps every core; not idempotent.	§3.3
Companion paper	<i>Pattern Arithmetic: Inference from Bowls and Paths</i> — reference atoms, classification, abstention; where unmixing and path reduction are left.	§1, §3.2, §12.1
Composition	How diverse units come together in one space: Bundle (presence, repetition dropped) and Collapse (every vector and count kept, any sorts); neither computes across the units; taken apart by Remove or UnCollapse; read as one unit through the derived pair.	§1.4
Contact point κ_{AB}	$\hat{n}_A + \hat{n}_B$, unnormalized; 0 at antipodal.	§2.9, eq. 14
Containment vector	For each vector, the position of its core in the atom: the grouping itself, part of the atom's value; not a meta vector.	§2.10, §2.11
Contract (what the algebra promises)	Determinism given the reading time; results seatable again; no source consumed, every result a new atom naming its operands; no atom edited; no verb moves a vector between cores. A use supplies seats, reading time, comparison rules, grain, resolution and the named conventions.	§1.4
Core	A group of vectors held together in an atom; any number of vectors; each vector is held by exactly one core. Reserved word.	§2.10
Costume	An illustration of the $\uplus$ -versus- T split (storm, table and chair, forest, pathogen); not a model.	§1.4, §9
Counting trap	$A \uplus A \uplus A \uplus A \setminus A = \emptyset$: four gathers are one seat, one removal empties it.	§2.5, eq. 7
Density	Many identities can share the stage without overwrite. Not a metric and not infinite information.	§2.4
Determinism	Same printed starting units (read at the same time where any carries an expiry) give the same result, step for step. Printing makes a result replayable; it does not make it right.	§1.4
Dictionary pin / doorbell	A seated unit for a name (man, storm, calm, Night). The algebra does not mint it; a lexicon or learning rule seats it.	§9.1, §5.2, §6.6
Direction $\hat{n}$	A pattern's semantic address, not its physical position.	§2.2, §10.1
Effective vector count Q_{eff}	$\mathcal{I}_{\text{str}}^2/\mathcal{I}_{\text{en}}$; how many equally strong vectors carry the same energy.	§2.10
Empty atom	Black: $N = 1$, $Q = 0$.	§2.3, §2.10
Empty bowl $\emptyset$	A bowl with no occupant; $\neq \mathbf{0}$. T_0 is defined, $T_\emptyset$ is not.	§2.5, eq. 6
Enclosure	Seats, rest and camps: where Assembly's separation comes from.	§10

Term	Meaning	Where
Energy (two senses)	The atom's information energy $\mathcal{I}_{\text{en}} = \sum r_i^2$ (§2.10, eq. 15); the interaction energy U_{AB} (§2.9, eq. 13). Not interchangeable.	§2.10, §2.9
Equality	Equal: the same vectors in the same grouping; meta vectors never enter. Exact.	§2.11
Expiry V_{expiry}	When a core stops being valid; absence = none; an expired core reads as black and may be removed by UnCollapse.	§2.11
F_i (force)	$-\nabla_{\vec{P}_i} \sum_{j \neq i} U_{ij}$.	§10.2, eq. 47
Freeze (at resolution η)	$\max_i F_i < \eta$. A freeze, not a stop.	§10.2, eq. 48
Grain	The partition of an atom's vectors into cores; UnCollapse acts along it; fixed at seating, coarsened by seating again, refined by no verb.	§3.3
Identity key	A core's name: the tuple of its sort, source link and timestamp, existing meta vectors; never enters equality.	§2.11, §3.4
Individual vs value	Under Collapse, Bundle, Mix the individual (with its meta vectors) persists; value is the vectors and their grouping.	§2.11
Informational vector	One of an atom's Q arrows $v_i = (r_i, \hat{n}_i, \chi_i)$, $0 < r_i \leq 1$.	§2.10
Levels are closed	A result may be seated again and Bundled, walked, read at the next level.	§3.1
E_L (lexicon)	A map from one language's tokens onto states; case and language do not move the address.	§6.6, §7, eq. 42
Link V_{link}	Backward provenance stored at birth; forward and bidirectional are views.	§2.11
Local position	A core's place in its atom's list of cores; the fallback when a key matches cores of different content.	§3.4
Location V_{location}	A point on a sphere carried as a meta vector; not a sort.	§2.11
Meta vector	A value that describes an atom and is no part of what it says.	§2.11
Mix $\boxplus$	Phasor reading of states of one sort sharing a direction; reads as the domain's own combination only where Axiom 0 admits the sort.	§2.6, §1.4
Mix across differing directions	Refused: a claim about Mix at one shared direction, not a permanent bar; a two-stage operator stays open.	§2.7
Mode (chladni)	A vibration mode; Mix exact within one, undefined across.	§2.7
Non-associativity (Mix)	A consequence of the cap: the binary operator is not associative; the one-step bowl reading is.	§2.6
Non-commutativity (T)	Inherited from $\text{SO}(3)$; the only thing that lets two poems from one bowl diverge.	§4.3, eq. 22
Occupant (register occupant)	The single current unit on the sphere that T acts on.	§2.2, §3.1
Offset	Copies a walk; does not mint the landing's name.	§5
Order gap	$ T_{w_2} T_{w_1} \hat{n}_x - T_{w_1} T_{w_2} \hat{n}_x $.	§4.4
Path / walk / trajectory	An ordered sequence of actors on an occupant; a poem is a T -walk. Order enters here and not in the bowl.	§3.1, §4.5
Phase χ	Internal phase / mood / hue: adds under T , sums as a phasor under Mix; reserved for mood, not language.	§2.2, §7
Polarity	$ r_A - r_B $.	§2.9
Pratyāhāra; vipratīṣedha	Pāṇini's sound-class shorthand and his conflict-resolution rule; offered as kinship, not identity.	§1
Reading	An operation that returns a state and leaves its inputs intact (Mix, a T -path, Collapse); also, expiry is a reading.	§2.6, §3.3, §2.11
Remove $\backslash$	Takes one member out of a bowl.	§2.5
Rest	A freeze at a stated resolution, never exact stillness.	§10.2
Result	Whatever the path makes of the bowl.	§3.1
Rotation	$\text{SO}(3)$ turn of $\hat{n}$ only.	§2.8
Same individual	Equal, and the meta vectors agree too: the stricter test ($==$ against "is").	§2.11
Seat $\vec{P}$	External position vector; a meta vector; enters only d_{AB} .	§10.1, eq. 46
Seated unit	A unit loaded on the sphere, pinned by a lexicon or a learned map.	§1.4, §3.1
Shell	(a) a width- w bin of strength in $\mathcal{I}_{\text{scn}}$ (§2.10); (b) one phased identity at a direction in Mix ("two shells, not an angle", §8.1).	§2.10, §8.1
Softening length ε_{AB}	$r_A + r_B$, a convention that reads r as a radius.	§2.9
Sort τ	Says which operators may legally act on a unit; does not move the arrow.	§2.2, §6

Term	Meaning	Where
Stage $\mathcal{S}$	The unit sphere (or its higher analogue): the sphere is the stage, the unit is the vector.	§2.2
Step δ	Numerical sampling convention for the descent.	§10.2
Strength r	Normalized activation in $[0, 1]$.	§2.2
Tape	An ordered sequence of tokens; a document.	§6.3
Teaching atom	One 4-tuple, $N = Q = 1$.	§2.2
Tier 0 / Tier 1	Inert seating; or explicit inclusion by a stated φ_τ .	§2.1
Timestamp $V_{\text{timestamp}}$	Absolute birth time of a core; a Collapse result has its own.	§2.11
Transform T	Rotate the occupant about the actor's axis by πr_w and add the actor's phase.	§4
Transport	Carries a tangent vector from seat m to seat w in Offset.	§5.1, eq. 33, 35
Turn (an atom)	Same rotation and phase shift on each vector; V_{derived} turns, scalar meta vectors are carried.	§2.11
UnCollapse $\ominus$	Takes out one core with its vectors, named by identity key or local position.	§3.4
UnMix $\boxminus$	Impossible as an inverse of the reading: sources not determined by the result; where they were kept, Remove or UnCollapse.	§3.2, §8.2
Valid (core)	At t iff t is earlier than every expiry it carries.	§2.11
Volume $\mathcal{I}_{\text{vol}}$	$\sum r_i^3$, "atom volume"; descriptive only, deliberately excluded from motion.	§2.10
White	Saturation of the bowl, or a full-spectrum name; never $r = \infty$.	§2.4
D	Distinct cores among the N .	§3.3
N	Cores.	§2.10
N_{dim}	Dimension of the ambient space; 3 is the standing assumption.	§4.4
Q	Informational vectors; "the atom's info number".	§2.10
U_{AB}	Assembly's interaction energy.	§2.9, eq. 13
V_{derived}	Uncapped sum of the informational vectors.	§2.10
z_B	Phasor sum of a bowl.	§2.6, eq. 9